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Upper-Limb Synergy Modeling from Real-World Contactless Motion Tracking and Validation via Robotic Hand Rendering

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***Abstract*— The extraction of low-dimensional kinematic synergies from movement data is foundational for understanding motor control and enabling intuitive human-robot interaction. However, conventional methods like Principal Component Analysis (PCA) are sensitive to the noise and artifacts inherent in practical motion capture. This study introduces Robust Principal Component Analysis (RPCA) as a superior alternative for synergy extraction from hand motion data. Analysis of hand motion data from 40 subjects across three cohorts (High, Medium, and Low-Quality) defined by landmark tracking quality revealed that PCA required 7–10 synergies to explain 95% of the variance, whereas RPCA achieved the same threshold with only 3–5 synergies, demonstrating a more compact representation. When dimensionality was fixed (k=4), RPCA consistently captured approximately 10% more variance than PCA and maintained stable performance as data quality degraded. Qualitatively, RPCA produced synergies with smoother temporal activation profiles and clearer separation between hand and arm dominant components, better reflecting known motor coordination patterns. RPCA also achieved lower normalized reconstruction error. The practical utility was validated by successfully mapping reconstructed postures to a 6-DoF anthropomorphic robotic hand. These results demonstrate that RPCA effectively isolates sparse tracking corruption to recover robust, low dimensional synergies, offering significant advantages for motor control analysis and the development of noise resilient human-machine interfaces.**

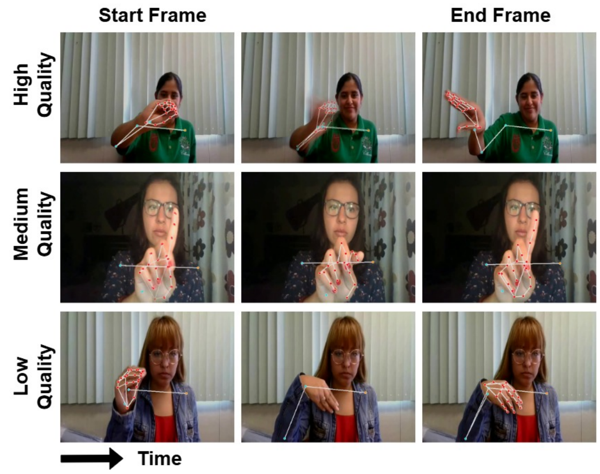

Fig. 1. Subjects grouped into three data quality cohorts based on landmark visibility is shown here. In the high-quality cohort, $> 95\%$ of landmarks were observed throughout the gesture execution (missing left elbow joint); in the medium-quality cohort, $> 90\%$ were observed (missing elbow joints); and in the low-quality cohort, $> 80\%$ were observed (missing hand joints). Representative examples include Sub 7 (high cohort) performing gesture G06 (rep 2), Sub 3 (medium cohort) performing gesture G01 (rep 3), and Sub 4 (low cohort) performing gesture G04 (rep 3).

I. Introduction

Upper-limb motor recovery following stroke or spinal cord injury is a complex rehabilitation challenge where successful outcomes depend not merely on restoring isolated joint movements, but rather on re-establishing the integrated, coordinated movement patterns across shoulder, elbow, wrist, and hand movement quality [1], [2]. Traditional assessment relies heavily on qualitative therapist observation using standardized clinical scales, which, while valuable, are subjective and lack sensitivity to subtle longitudinal changes. Alternatively, laboratory-based motion capture systems offers high fidelity kinematic data but are costly, environment-dependent, and impractical for frequent home-based monitoring. In contrast, vision-based tracking systems using consumer-grade cameras and deep learning estimators promises scalable, markerless motion analysis deployable in clinics and homes [3]. However, this promise is currently undermined by a fundamental reliability problem: these vision systems suffer from persistent landmark jitters, temporary dropout during self-occlusion or rapid motion, and artifacts induced by challenging lighting or backgrounds which are challenges that are amplified in unstructured home environments where such technology is most needed [4].

Current approaches typically address tracking reliability and motor analysis as a sequential problem: first, apply various filters or imputations to clean the noisy kinematic signal, and then extract clinically relevant features. We propose a paradigm shift by arguing that robustness is embedded within the movement representation itself, not applied as a preliminary preprocessing step [5], [6]. This view is strongly motivated by foundational motor control literature, which reveals that the apparent complexity of human movement emerges from the flexible combination of a small set of low-dimensional building blocks called synergies – coordinated patterns of muscle or joint activation that reflect the nervous system's strategy for simplifying motor control [7]. These synergies provide not only an efficient, compact representation of movement but also a physiologically interpretable framework for understanding motor synergies, including the compensatory movements that are critical markers of impairment and recovery [8].

In this study, we hypothesize that extracting kinematic

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synergies using methods explicitly robust to sparse outliers can inherently filter tracking artifacts while preserving clinically relevant coordination patterns. Furthermore, we assert that these extracted synergies are not merely analytical constructs but can be directly translated into executable motion primitives for a robot, thereby creating a valuable closed-loop pipeline from assessment to robotic assistance. This integration offers a pathway towards low-cost rehabilitation systems in which a robot can assess patient movements via a simple camera, identify coordination deficits through synergy analysis, and guide corrective movement within a single computational framework.

II. Related Works

The concept of kinematic synergies provides a compact representation of coordinated joint motion, offering insights into the underlying organizational principles of the motor system [9], [10]. In neurorehabilitation, analyzing changes in synergy composition, timing and dimensionality enables quantitative assessment of movement quality, neurological recovery, and therapeutic efficacy [11]. This framework has also informed rehabilitation robotics, with exoskeleton controllers designed to assist along synergy directions to promote naturalistic motions [12], and robotic assessments leveraging synergy similarity metrics to evaluate how well a patient's movement aligns with healthy patterns [13]. Synergy-based control has further enabled low-dimensional assistance in upper-limb exoskeletons [14], and dexterous prosthetic hands [15]. However, these approaches universally rely on clean, laboratory-grade kinematic data from motion capture or instrumented gloves, leaving the challenge of extracting reliable synergies from noisy, and incomplete consumer vision data largely unaddressed.

Advances in consumer-grade cameras and deep learning have accelerated interest in vision based analysis. Pose estimation frameworks such as OpenPose and MediaPipe enable real-time estimation of body and hand kinematics from standard video, supporting applications in rehabilitation, assisted living, sports science, and clinical assessment [16], [17]. While preliminary studies have demonstrated feasibility in stroke [18] and Parkinson's disease assessment [19], clinical adoption remains limited by sensitivity to occlusions, motion blur, lighting variability conditions, leading to landmark dropout, jitter, and physiologically impossible joint angles [4]. Existing mitigation strategies typically treat these issues as a preprocessing problem, applying temporal filters, kinematic smoothers or biomechanical constraints to clean the landmarks before analysis [5], [6]. In contrast, our work proposes embedding robustness directly into synergy extraction process.

RPCA decomposes data into a low-rank structure and a sparse corruption [20] and has been widely used for video background subtraction [21], fault detection and biomechanical denoising [22], [23]. Rather than applying RPCA solely as a signal leaning step, we leverage it to derive synergy bases from the low-rank component of noisy vision-based kinematics, extracting synergies that are less contaminated by tracking artifacts and more representative of underlying motor coordination.

Transferring human motion to robots is a long-standing challenge involving kinematic mismatches, stability constraints, and noisy demonstrations [24]. Synergy-based representations offer an effective means of reducing motion dimensionality, making it easier to learn, generalize and reproduce by robots [10], but existing methods assume high-quality motion capture data [25]. By addressing these realistic scenario of translating noisy, incomplete, vision-based human motion into stable robot execution, this work advances the development of a low-cost, vision-driven rehabilitation system capable of learning from demonstrations captured in everyday environments.

III. Methods

A. Dataset

To evaluate the robustness of the proposed synergy extraction pipeline under realistic vision-tracking constraints, we use the publicly available IPN Hand dataset [26]. The dataset comprises of 50 subjects performing 13 dynamic hand gestures (B0A, B0B, G01 to G11) recorded across diverse environments. Unlike datasets commonly used in synergy extraction studies, IPN Hand does not contain controlled, discrete movement cycles. In laboratory settings, gestures are typically performed from a predefined rest posture with clear segmentation, resulting in clean and repeated motion primitives. In contrast, the IPN Hand captures continuous, naturalistic gestures performed without explicit starting poses or transitional pauses introducing significant intra-class variation in movement onset, speed, and trajectory which are challenges of real-world scenarios where a vision system must interpret movement from an arbitrary ongoing state 1. The dataset intentionally incorporates real-world adversities such as cluttered backgrounds, extreme illumination conditions, and both static and dynamic environments, stressing the capabilities of vision-based trackers. Videos were recorded using subjects' personal devices, leading to natural variations in camera placement and subject-to-camera distance. All recordings were provided as 640×480 RGB video streams at 30 fps.

B. Preprocessing

Each video contains a subject continuously performing 21 gestures with three random breaks. Using the provided annotation files, video frames were temporally segmented to isolate individual gesture instances. As a result, each subject contributes 13 gesture classes with four repetitions per gesture, except for B0A and B0B (reps > 4). For each video segment, we apply a custom pipeline combining MediaPipe Hands [17] and Blaze Pose [27] to extract a total of 26 anatomical landmarks per frame: 5 for the upper limb (left and right shoulder joints, left and right elbow joints, left wrist joint) and 21 for the hand (wrist and finger joints similar to the standard MediaPipe hand topology). Each clip is temporally resampled fixed length of $N = 60$ frames to ensure consistency across gestures of varying durations. The resulting data was organized into a 6D tensor represented as $X \in \mathbb{R}^{(S\times T\times R\times L\times C\times N)}$, where S denotes the number of subjects, T is the number of gestures captured, R indicates the number of repetitions per task, L represents the landmarks, C corresponds to the spatial coordinates (x,y,z) and N is the number of samples per trial. To systematically study robustness to tracking degradation, subjects were stratified into three cohorts based on the proportion of frames in which all landmarks were successfully detected over the 60 frame window: the High-Quality cohort ($\geq$ 95% non-NaN frames across all landmarks, $n = 20$), the Moderate-Quality cohort (90-95% completeness, $n = 10$), and the Low-Quality cohort (80-90% completeness with observable noise, $n = 10$). The tracking quality of the landmarks can be observed in Fig. 1. This stratification enables controlled evaluation as input quality progressively degrades.

C. Synergy Extraction Models

We compare two models for extracting low-dimensional synergy representations from high-dimensional landmark data. To evaluate generalization, training and test sets are designed to be kinematically distinct. For each subject, the first 10 gestures are used to learn the synergy basis, while the remaining three gestures are held out for evaluation. Training data are reshaped into a 2D matrix $\mathbf{X}_{\text{train}} \in \mathbb{R}^{F\times M}$, where $F = L\times C$ features per frame, and M is the total number of concatenated frames across selected gestures and repetitions. The test matrix X_{test} is constructed similarly. All features are standardized using the mean and standard deviation computed from the training set.

PCA serves as the baseline synergy extraction method, consistent with our prior work [10]. Singular Value Decomposition (SVD) is applied to the standardized training matrix: $\mathbf{X_{train}} = \mathbf{U}\Sigma\mathbf{V}^T$. The columns of U represents synergy directions ordered by decreasing variance, as determined by the singular values in Σ. The top k synergies, denoted U_k, are retained such that they capture at least 95% of the cumulative variance. Test data reconstruction is obtained through projection: $\hat{\mathbf{X}}_{\text{test}} = \mathbf{U}_k\mathbf{U}_k^T\mathbf{X}_{\text{test}}$. While PCA is optimal under Gaussian noise assumptions, it is highly sensitive to sparse, large magnitude outliers, which commonly arise from vision tracking failures.

To address this limitation, we employed RPCA using Principal Component Pursuit formulation: $\min_{L,S} \|L\|_* + \lambda\|S\|_1$ subject to $X_{train} = L+S$, where $\|L\|_*$is the nuclear norm promoting low rank, $\|S\|_1$ enforces sparsity, and λ balances the two terms [20]. This decomposition separates structured motion (L) from sparse corruption (S), including dropout-induced spikes and tracking artifacts. The optimization is solved using an Inexact Augmented Lagrange Multiplier (IALM) method [23]. SVD is subsequently applied to L, and the top k eigenvectors form the synergy basis U_k. Test data reconstruction follows the same projection formulation as PCA, with the critical distinction that the synergy basis is learned from the low-rank component rather than raw data.

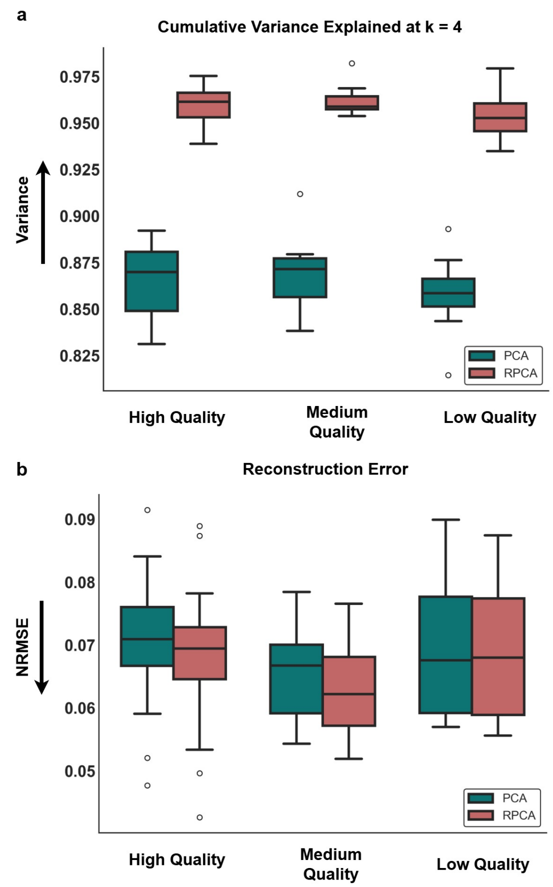

Fig. 2. Cumulative explained variance and reconstruction error at $k = 4$ for PCA and RPCA across all three data-quality cohorts. (**a**) Cumulative variance, showing consistently higher variance captured by RPCA compared to PCA. (**b**) Reconstruction error (NRMSE), where RPCA achieves lower error across cohorts. Green denotes PCA and red represents RPCA.

D. Evaluation Metrics

A multi-criteria evaluation strategy is used to assess the kinematic reconstruction and representation quality. Reconstruction accuracy was computed using Normalized Root Mean Square Error (NRMSE) between the original held-out test data X_{test} and its reconstruction $\hat{X}_{\text{test}}$, $NRMSE = \frac{\|X_{test}-\hat{X}_{test}\|_F}{\|X_{test}\|_F}$, where $\|\cdot\|_F$ is the Frobenius norm. Representation quality is assessed via explained variance, defined as the fraction of the total variance captured by the retained top k synergies $\left(\frac{\sum_{i=1}^{k}\sigma_i^2}{\sum_{i=1}^{r}\sigma_i^2}\right)$, where σ_i are the singular values and r is the rank.

To support clinical interpretability, we introduce two visualization tools:(i) spatial heatmaps that color-code each landmark to its weight in a synergy vector and (ii) temporal activation plots synergy activations at $a(t) = U_k^T x(t)$ over time, with variance contributions decomposed into arm and hand components.

IV. Results

A. Synergy Dimensionality and Robustness

A consistent and central observation after applying PCA and RPCA to extract kinematic synergies from different

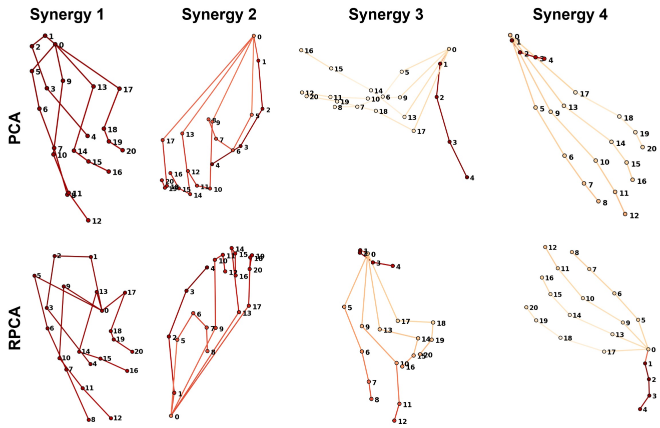

Fig. 3. First four synergies extracted by PCA (top row) and RPCA (bottom row) for Subject 5 (Medium Cohort) is shown here. Hand landmarks are shown as black dots; connecting lines are colored by each landmark's contribution to the synergy, with red indicating highest and yellow lowest. Here 0-Wrist, 1-Thumb_CMC, 2-Thumb_MCP, 3-Thumb_IP, 4-Thumb_Tip, 5-Index_MCP, 6-Index_PIP, 7-Index_DIP,8-Index_Tip, 9-Middle_MCP, 10-Middle_PIP, 11-Middle_DIP, 12-Middle_Tip, 13-Ring_MCP, 14-Ring_PIP, 15-Ring_DIP, 16-Ring_Tip,17-Pinky_MCP, 18-Pinky_PIP, 19-Pinky_DIP, 20-Pinky_Tip

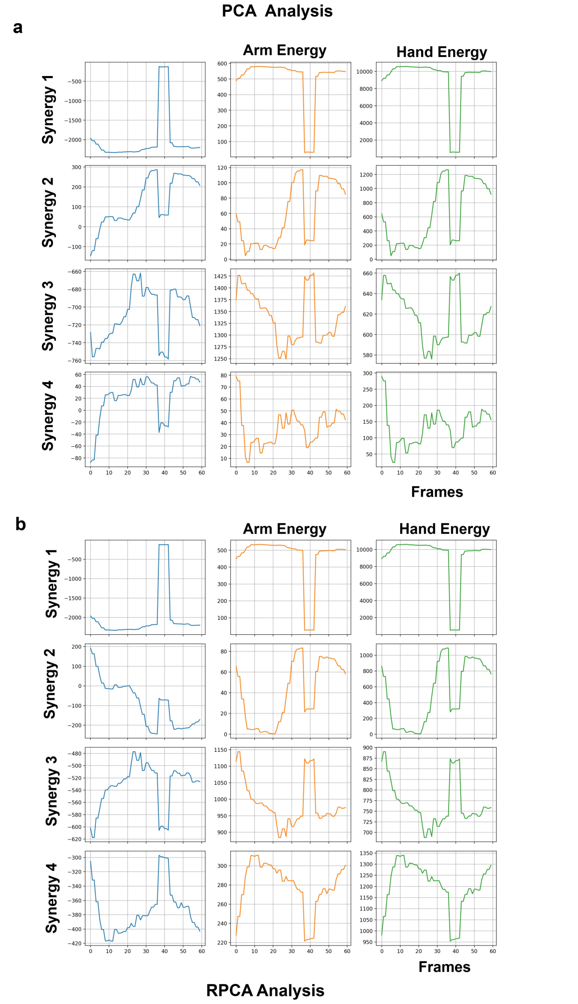

Fig. 4. Synergy activations, arm energy, and hand energy over 60 frames when reconstructing the test task 1, rep 1 from Sub 5 (Medium-Quality cohort) with PCA (**a**) and RPCA (**b**) is illustrated here. Note the smoother temporal profiles produced by RPCA compared to PCA.

quality cohort was that RPCA yielded a markedly more compact representation of coordination motion than PCA. Across three cohorts totaling 40 subjects, PCA typically required 7-10 synergies to reach the 95% variance threshold, indicating sensitivity to distributed noise and tracking artifacts. In contrast, RPCA achieved the same 95% threshold with only 3-5 synergies for all subjects, demonstrating its ability to isolate sparse corruption and concentrate structured motion into a lower-dimensional subspace.

To enable capacity-matched, all subsequent analyses fixed the number of synergies to $k = 4$ for both methods. Under this constraint, RPCA consistently captured approximately 10% more variance than PCA across all cohorts as shown in Fig. 2(a). Paired statistical tests revealed highly significant differences in explained variance favoring RPCA for High, Moderate, and Low-Quality cohorts (all $p < 10^{-7}$). Importantly, PCA's explained variance degraded as landmark quality worsened, whereas RPCA maintained stable variance capture regardless of data fidelity, with lower inter-subject variability. In the Low-Quality cohort, PCA's explained variance with $k = 4$ dropped as low as 85% for individual subjects, while RPCA consistently reaching 95% or more, highlighting its robustness under severe tracking corruptions. The first four kinematic synergies extracted by PCA and RPCA for Subject 5, Medium-Quality cohort are visualized in Fig. 3. The synergy contributions are color-coded from red (highest contribution) to yellow (lowest contribution).

B. Spatial and Temporal Structure of Synergies

Qualitative inspection of the temporal activation profiles and corresponding arm and hand energy distributions reveals pronounced differences between PCA and RPCA derived synergies (Fig. 4). In the PCA case, synergy activations exhibit sharp discontinuities and abrupt sign changes, accompanied by simultaneous collapses in both arm and hand energy. These impulsive drops, most evident around frame 40, coincide with landmark dropout events and indicate that PCA directly propagates transient tracking failures into the synergy activations. As a result, arm and hand dominant components frequently activate concurrently, leading to overlapping temporal profiles and reduced interpretability of the underlying coordination structure.

In contrast, RPCA produces markedly smoother and more temporally coherent activation profiles. Although the same landmark loss event is present, its effect is largely absorbed by the sparse component, allowing the low-rank synergies to maintain stable activation and energy distributions. Hand dominant synergies consistently activate earlier in the movement, while arm dominant synergies peak later during the transport phase, reflecting a physiologically plausible sequencing of grasp preparation followed by arm motion. Importantly, arm and hand energy remain well separated over time, with minimal overlap and no spurious collapses. Thus RPCA preserves meaningful temporal coordination patterns

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while suppressing noise-induced artifacts, enabling more reliable interpretation of vision-based kinematic synergies.

C. Reconstruction Performance and Robot Validation

RPCA's ability to extract a more informative set of synergies did not come at the cost of reconstruction accuracy. Across all cohorts, under the constraint of $k = 4$, RPCA achieved lower NRMSE compared to PCA (Fig. 2(b)). This superior robustness of RPCA was most evident under dimensional constraint. While PCA initially showed lower NRMSE when allowed higher dimensionality ($k > 7$), its performance significantly degraded when restricted to the same four components required by RPCA. Specifically, in the High-Quality cohort, PCA's mean NRMSE rose to 0.0705, whereas RPCA maintained a superior reconstruction error of 0.0680. This trend persisted in Medium-Quality cohort as well. The trend was more pronounced in the Low-Quality cohort, where RPCA's NRMSE remained stable (0.0681 ± 0.011) despite increased tracking noise, while PCA displayed higher sensitivity to artifacts, an increase in NRMSE when using $k > 7$. These results suggest that RPCA not only captures more variance with fewer components but also produces a more faithful and robust reconstruction of the underlying movement kinematics by effectively filtering out non-Gaussian noise during the decomposition process.

To further validate the practical utility of the extracted synergies, reconstructed hand postures were transferred to a 6-DoF anthropomorphic robotic hand (Inspire Hand, Cogito Instruments, China). This allowed qualitative assessment of end postures resulting from each decomposition method. Fig. 5 shows the reconstruction of Subject 13 (High-Quality cohort) performing test gesture 1, rep 3. A percentile-based calibration was applied to the robot's actuation range $[1, 1000]$, and control was implemented in Python via TCP communication. The RPCA-based reconstruction produced more stable postures as shown in Fig. 5(d).

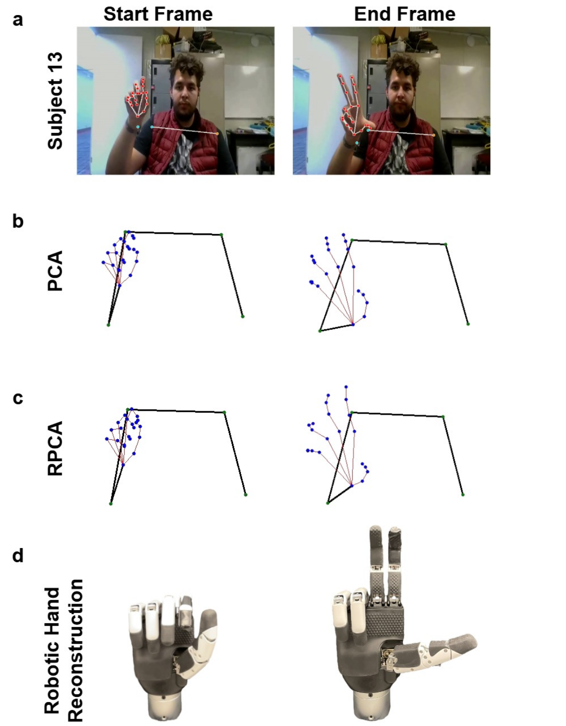

Fig. 5. Subject 13, performing test gesture 1 (G10), rep 3 is illustrated here. Mediapipe applied landmarks are shown in (a). Reconstruction of the movement using PCA and RPCA on a skeletal framework is shown in (b) and (c). (d) represents the start and end frames gesture postures of RPCA on the robotic hand.

V. Discussion

This study investigated the application of RPCA for extracting kinematic synergies from hand motion data with varying levels of tracking fidelity. Our key findings indicate that RPCA provides a more compact, robust, and physiologically plausible representation of movement synergies compared to conventional PCA, particularly in the presence of sparse noise and artifacts common in markerless motion capture.

A. Interpretation of Principal Findings

The observation that RPCA consistently captured 95% of the variance with 3–5 synergies, roughly half the number required by PCA, aligns with its mathematical foundation. PCA assumes Gaussian noise distributed across all measurements, causing outliers and tracking dropouts to inflate the perceived dimensionality of the data. In contrast, RPCA explicitly models the data as a sum of a low-rank matrix (structured motion) and a sparse matrix (corruptions). This allows RPCA to isolate non-Gaussian artifacts such as landmark jitter and temporary dropouts, thereby recovering a lower dimensional subspace that more accurately reflects the true biomechanical degrees of freedom being coordinated. Our results confirm that RPCA is not merely a denoising filter; it fundamentally alters the extracted basis set, concentrating variance into fewer, more meaningful components.

A significant advantage of RPCA was its consistent performance across High, Medium, and Low-Quality cohorts. While PCA's explained variance degraded with worsening data quality, RPCA maintained stable variance capture and lower reconstruction error (Fig. 2). This robustness is critical for real world applications where motion capture conditions are rarely ideal. The ability to derive reliable synergies from noisy, ”in-the-wild” data expands the potential use of synergy analysis in clinical, ergonomic, and recreational settings where high-fidelity laboratory capture is impractical.

Beyond quantitative metrics, the qualitative differences in synergy structure are evident. The smoother RPCA activations are not the result of temporal filtering, but may emerge directly from the decomposition itself, underscoring the advantage of embedding robustness within the representation rather than applying post hoc smoothing. The observed hand synergies activating earlier during gesture formation, followed by arm synergies for transport mirrors known motor control strategies for reach-to-grasp movements [9]. This suggests that RPCA is better at recovering the underlying neural control structure by filtering out corruptions that disrupt temporal coherence in PCA. The spatial organization

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of RPCA synergies also appeared less contaminated by outlier landmarks, leading to more interpretable patterns of joint coupling.

B. Implications for Motor Control and Rehabilitation Robotics

From a motor control perspective, the low-dimensional subspaces extracted by RPCA may offer a clearer window into the motor modules used by the central nervous system. The stability of these subspaces across noise conditions suggests that RPCA could be a valuable tool for identifying pathological movement patterns in clinical populations, where movement variability and measurement noise are often high. The successful transfer of RPCA reconstructed postures to an anthropomorphic robotic hand demonstrates the method's utility for human-robot interaction. The more accurate and noise resistant synergy set provided by RPCA can lead to more natural and intuitive prosthesis control schemes, as it better represents the user's intended motion. Furthermore, the lower dimensionality reduces the complexity of control mapping, potentially easing user learning and adaptation.

VI. Conclusion

Findings from this study shift the paradigm for analyzing movement in real-world settings. Instead of treating robustness as a preprocessing step, we integrate it into the core representation-learning process. This approach paves the way for reliable, low-cost motor assessment outside the laboratory and supports the development of adaptive human-robot interfaces that can interpret movement intention even amidst the noise of everyday environments. Future work will focus on using time variant synergy implementations and clinical validation to translate this methodological advance into tangible tools for rehabilitation and assistive technology.

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